



Enterprise Asset Management (EAMS) v15

Asset Assessments

End User Documentation

Update 3/1/2024

Asset Assessments

MaintStar can record, track and report an Asset's recorded Assessment. Assessments allow for an organization to provide Asset condition ratings/assessments to report on its condition. As each Asset may be comprised of different ratings, MaintStar allows for a multitude of Rating Types to suit each need. For instance, Road Segments are often rated using a Pavement Condition Index (PCI) of 0-100 whereas other Assets may be a simple 1-5 or A-F scale. These rating types are set at the Asset Type level, which is further discussed below.

If Assessments are to be used, 3 critical concepts must be understood.

- **Condition Index:** this is the type of condition to be applied, for instance "PCI" of 0-100 with 0 representing Failed and 100 representing Excellent condition
- **Condition Rating:** the actual scale of which the Index uses – for instance:
 - Vehicle Rating (Condition Index scale 1-5)
 - 1 = Failed
 - 2 = Poor
 - 3 = Fair
 - 4 = Good
 - 5 = Excellent
- **Asset Type Condition Index:** the condition index can be set at the Asset Type level so that all Assets inherit the index

When recording an Assessment, the following information can be tracked to help determine an asset's condition:

- Condition Index and Rating
- Assessed Value
- Assessed Replacement Cost
- Expected Life
- Remaining Life
- Observed Issues/Conditions as comments

Assessment Setup

As mentioned above, several things must be in place to utilize Asset Assessments: Condition Indexes, Ratings and the Asset Type must be configured.

NOTE: The following requires appropriate permissions.

Condition Index

Set the Condition Index by going to Setup > Tables > Condition Index.

- Provide Index Name
- Provide the Worst and Best Rating Limits – this provides the boundaries of the scale, ie "1 is best, 5 is worst"

Tables

Table list: Condition Index

Condition Indexes, such as PCI, which are used to measure the condition of an Asset.

Add

	Condition Index	Description	Best Rating	Worst Rating
☐ Edit	1 TO 10	1 is best, 10 is worst	1.0	10.0
☐ Edit	1 TO 5	1 is best, 5 is worst	1.0	5.0
☐ Edit	Alligator	Alligator cracking in asphalt	1.0	10.0
☐ Edit	Bridge-BELA	Elastomeric bearing pad	1.0	4.0
☐ Edit	Bridge-CABW	Concrete-abutment and wingwalls	1.0	4.0
☐ Edit	Bridge-CDSL	Concrete-deck slab	1.0	4.0
☐ Edit	Bridge-CPHS	Concrete-pier headstock	1.0	4.0
☐ Edit	Bridge-CPIL	Concrete-piles/piers	1.0	4.0
☐ Edit	Bridge-CPRG	Concrete-pre-tensioned girder	1.0	4.0
☐ Edit	Bridge-JNOS	Joint-no seal	1.0	4.0
☐ Edit	Bridge-JPOS	Pourable/Cork joint seal	1.0	4.0
☐ Edit	Bridge-MAPP	Approach carriageway	1.0	4.0
☐ Edit	Bridge-MBAT	Batter protection	1.0	4.0
☐ Edit	Bridge-MGCL	General cleaning	1.0	4.0
☐ Edit	Bridge-MWES	Wearing surface	1.0	4.0
☐ Edit	Bridge-MWWY	Waterway	1.0	4.0
☐ Edit	Bridge-RMET	Metal railing	1.0	4.0
☐ Edit	Bridge-RMIS	Miscellaneous railing	1.0	4.0
☐ Edit	Bridge-RPNT	Railing paint work	1.0	4.0
☐ Edit	Bridge-UCPL	Underwater CPIL Concrete-Pile	1.0	4.0

Row Count: 36

Condition Rating

Set the Condition Rating by going to Setup > Tables > Condition Rating. The Condition Index used above will be used for instructional purposes.

- Click Add to add a new rating (step will be completed for each rating)
- Select the index created above
- Create ratings from 1 to 5 with corresponding rating values

Tables

Table list: Condition Rating

Possible values for Condition Rating of an Asset, for a given Condition Index

Add

	Condition Index	Condition Rating	Description
☐ Edit	pci		
☐ Edit	PCI	0.0	Failed
☐ Edit	PCI	10.0	Poor
☐ Edit	PCI	20.0	Poor
☐ Edit	PCI	30.0	Fair to Poor
☐ Edit	PCI	40.0	Fair to Poor
☐ Edit	PCI	50.0	Fair
☐ Edit	PCI	60.0	Fair
☐ Edit	PCI	70.0	Good
☐ Edit	PCI	80.0	Good condition
☐ Edit	PCI	90.0	Excellent
☐ Edit	PCI	100.0	Excellent condition

Row Count: 11

Asset Type Setup

Configure the Asset Type Condition Index by navigating to Assets > Asset Type. Locate the desired Asset Type for which the Index will be applied. Click on the “Condition Index” tab and click Add. Add the desired Condition Indexes. Set the “Default” flag to the appropriate Condition Index.

Asset Type	Description	Active	Equipment	Vehicle	Workgroup	Customization Class	Meter Type 1	Meter Type 2	Meter Type 3	Default COF	Parent Asset Type	Bill Rate	Mfg Estimated Lifespan	SVF Flag
MAINTAINED ROAD	MAINTAINED ROAD	☒	☐	☐	HIGHWAY	HIGHWAY STRUCTURES	COUNTS					0.00	0	☐

Asset Type	Condition Index	Default	Description	Best Rating	Worst Rating	Add By	Add Date	Upd By	Upd Date
MAINTAINED ROAD	PCI	☒	Pavement Condition Index	100.0	0.0	Admin	03/06/2024		

NOTE: Multiple Indexes can be applied but only ONE of them is considered a Default index.

NOTE: Even though the setup is complete, Assets will only receive an Assessment once one is recorded.

Recording an Assessment

Recording an Assessment against an Asset can be completed on the Asset itself. To do so:

- Go to Assets > Assets
- Locate the desired Asset
- Click the Assessments Tab
- Click Add, select the appropriate Condition Index
- Record the Rating and complete desired Assessment Values

Viewing an Assets Historic Assessment Data

Historic Asset Assessments can be viewed directly from the Asset itself. This can be useful for analyzing asset condition over a period of time.

- Go to Assets > Assets
- Locate the desired Asset
- Click the Assessments Tab

Asset ID	Asset Type	Description
1	HIGHWAY SIGN	1 - Territorial
100	HIGHWAY SIGN	100 - Highw (Arizona Dep Transp
1000	HIGHWAY SIGN	1000 - White Lake Road (/
10001	HIGHWAY	10001 - Rust

Date	Assessed By	Condition Index	Condition Rating	Condition Descr	Other Not
04/02/2024	Admin	Daytime Sign	3.0	Fair	
03/27/2024	Admin	NRRT Rating	1.0	Pass	

Creating an Assessment Query

Utilizing the Query module to view all Asset Assessment data can be helpful to analyze a collection of asset conditions. With this data an organization can make data based decisions for replacement, budgeting or planning purposes.

NOTE: The following requires appropriate permissions.

- Go to Reports > Query
- Select New Query
- Provide a name for the query
- Set the table to: asset_assessments
- Add the desired fields, the following are minimum recommendations:
 - Asset ID
 - Asset Type
 - Condition Index
 - Condition Rating
 - Condition Description
 - Assessment Date

Additional Assessment Criteria

To truly take advantage of Asset Assessments beyond simple ratings, MaintStar recommends that Assets contain other key information. Adding this data will help to further assess assets based on cost, expected life, etc. As such, the following fields are recommended:

- **Original Value:** the original cost/value of the Asset
- **Install Date:** this date is used to indicate when the asset becomes active – not to be confused with Date in Service, Date Built, or Purchase Date
- **Replacement Cost:** the anticipated cost to replace the asset
- **Expected Life in Years:** set at Asset Type level

As this data is entered and Assets recorded, additional analysis can be completed through reporting or the optional (licensed) Asset Management module.